

JUSTIN SEJIN PARK

justinpark0902@gmail.com • 510.513.5799 • linkedin.com/in/justinpark-eng

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Master of Science in Mechanical Engineering | Expected: May 2027

Bachelor of Science in Mechanical Engineering, Highest Honors | May 2026 | GPA 3.96/4.0

ENGINEERING & TECHNICAL EXPERIENCE

Multirotors Engineering Intern - Heven Aerotech

06/2026 – 07/2026

- Developed a custom Python-based drone performance modeling tool to calculate flight time and range using payload weight, airspeed, altitude, battery capacity, and thrust data.
- Reverse-engineered an existing Group 2 multirotor platform, debugging and documenting all subsystem integration including flight control and avionics, companion computer, radio communication links, and video.
- Executed system integration and thrust testing, authoring comprehensive documentation on configuring flight software, establishing remote ethernet-based telemetry networks, and setting up video feeds.

Height-Adjustable Stabilized Camera Rig for Robodogs, Capstone Project

08/2025 – 12/2025

- Developed mechanical design of a stabilized camera rig for a Unitree Go2 Pro robot, enabling the capture of distortion-free images required for accurate 3D mapping.
- Designed a height-locking system for adjustability between 4 - 8 feet and 2-axis gimbal in Solidworks, using its integrated FEA tools for rapid concept evaluation before detailed analysis.
- Integrated an IMU and brushless motors with an STM32-based microcontroller to form an active 2-axis gimbal.

Independent Project: Large-scale Autonomous Land Survey Aircraft

01/2024 – Present

- Fully integrated Pixhawk flight controllers, avionics, powertrain, and tuned flight dynamics using PX4 firmware and QGroundControl.
- Solo-designed and fabricated all mechanical components, including airframe, suspension landing gear, electronics mounting, under limited budget.
- Iterated through four unique airframe designs with wingspans ranging from 65-85 inches, validated through extensive field testing.

Airframe & Avionics Engineer, Experimental Flights VIP - Georgia Tech

08/2024 – 12/2024

- Acted as the technical lead for a VTOL subteam, guiding the conceptualization and structural design of a custom tri-rotor tiltrotor fixed-wing airframe.
 - Guided the complete Pixhawk avionics integration process, bridging mechanical airframe design with onboard flight control systems and sensor payloads.
-

ACADEMIC & TEACHING EXPERIENCE

ME 4853 Graduate Teaching Assistant – Georgia Tech

08/2026 - Present

- Guiding interdisciplinary student teams through the product realization cycle, providing technical support from initial concept generation to physical prototyping.
- Instructing students on design engineering tool implementation, including simulation-based design and low-volume prototyping techniques.

ME 2110 Undergraduate Laboratory Supervising Grader – Georgia Tech

01/2025 - 05/2026

- Managed competition logistics and conducted rapid, pre-match field resets and safety checks, helping to maintain a consistent and fair playing environment.
 - Instructed students on technical fabrication methods, delivering clear, safety-focused training on the proper use of lab machinery (lathes, mills, laser cutters, and 3D printers).
-

TECHNICAL SKILLS

Software & Hardware: Python, SolidWorks, SolidWorks Simulation, Onshape, Matlab/Simulink, LaTeX, UAV firmware and GCS (PX4, ArduPilot, Mission Planner, QGroundControl), Raspberry Pi, ESP32